

Introduction to Data Science and Engineering

- Data visualization principles



Some materials courtesy of
Rafael A. Irizarry, and are modified
from the original version.

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Slide deck originally created by RB Luo



In this lecture

- How to do a good visualization? /
How to avoid bad visualization?
- Case study: vaccines and measles



In this lecture

- **How to do a good visualization? /
How to avoid bad visualization?**
- Case study: vaccines and measles



How to do good data visualization?

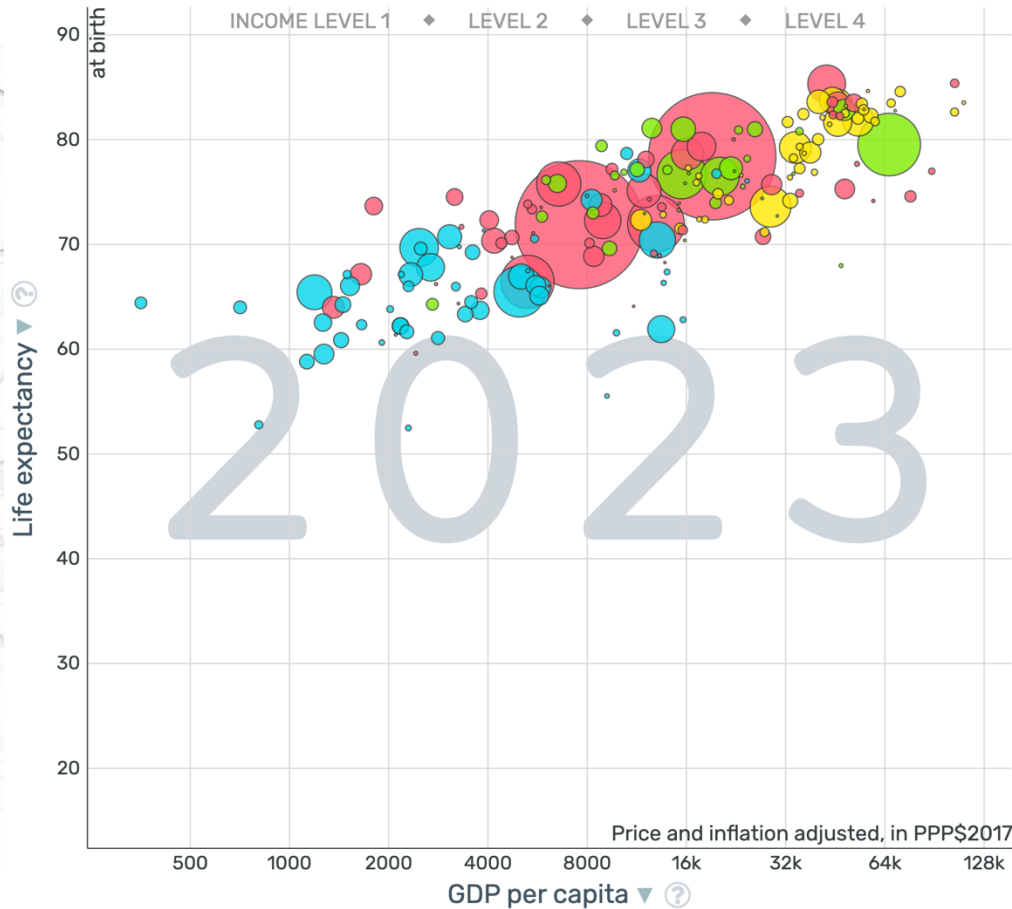
We have demonstrated the usefulness and power of data visualization through many case studies.

When we want to use data visualization to communicate ideas in the future, what will be the best practice? And reversely, what are some bad practice that we shall avoid?

Principle #1: Always keep in mind the primary message of your plot.

- Showing/comparing some quantities
- Showing/comparing distributions
- Showing relationship between variables

What can we use for visualization?



X-axis:

position / length

Y-axis:

position / length

Movement

Shape:

O or X

Size:

o or O

Angle/orientation:

↑ or →

Patterns/line styles

Color (hue):

red or blue

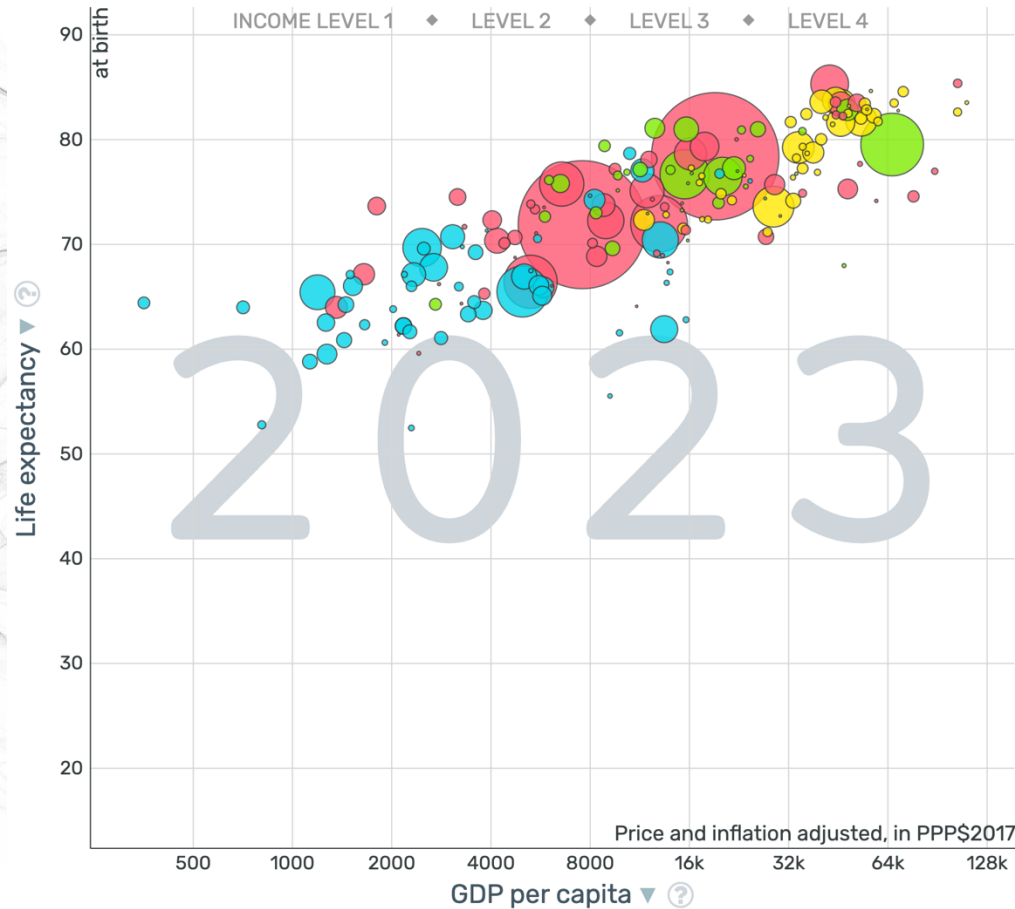
Color (shade):

light or dark

Color (opacity):

opaque or trans

What can we use for visualization?



For numerical variables:

X/Y (position/length), motion >>

size, angle/orientation >

color (shade), color(opacity)

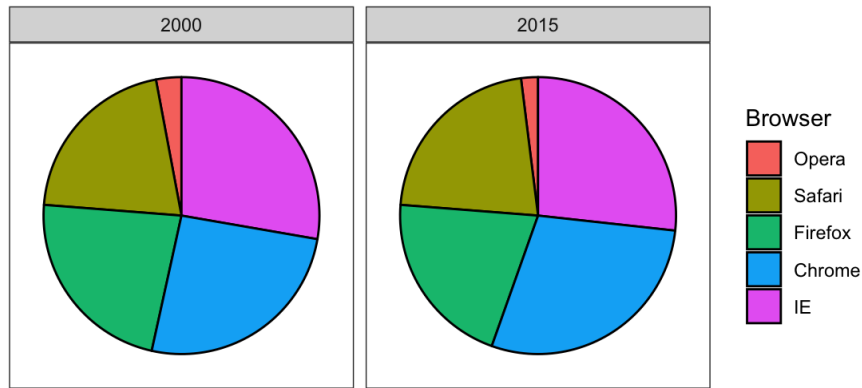
For categorical variables:

X/Y (position/length) >

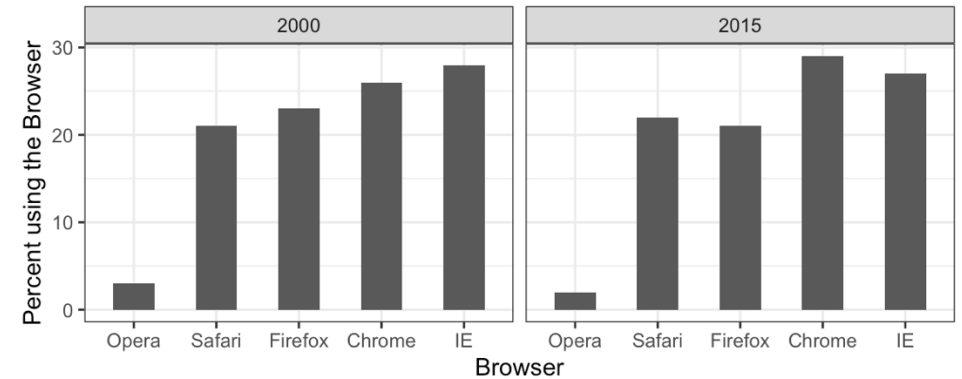
Shape, patterns/line styles, color(hue) >

Size, angle/orientation, color (shade)

Why pie chart should be avoided?



vs.



For numerical variables:

X/Y (position/length), motion >>

size, angle/orientation >

color (shade), color(opacity)

Note

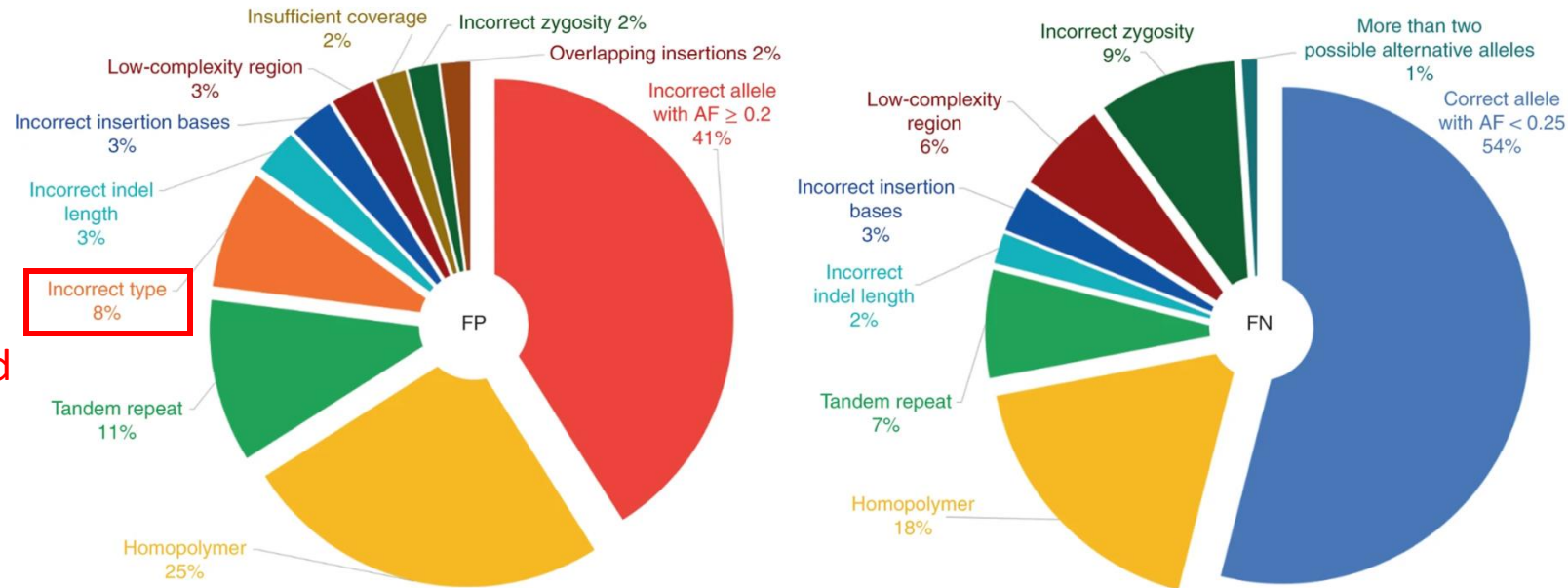
Pie charts are a very bad way of displaying information. The eye is good at judging linear measures and bad at judging relative areas. A bar chart or dot chart is a preferable way of displaying this type of data.

Cleveland (1985), page 264: "Data that can be shown by pie charts always can be shown by a dot chart. This means that judgements of position along a common scale can be made instead of the less accurate angle judgements." This statement is based on the empirical investigations of Cleveland and McGill as well as investigations by perceptual psychologists.

If your client insists

Fig. 3: The category distribution of FPs and FNs made by Clair in the 1:168×|2:64× experiment on ONT data and six genome browser screen captures showing examples of different categories.

From: [Exploring the limit of using a deep neural network on pileup data for germline variant calling](#)



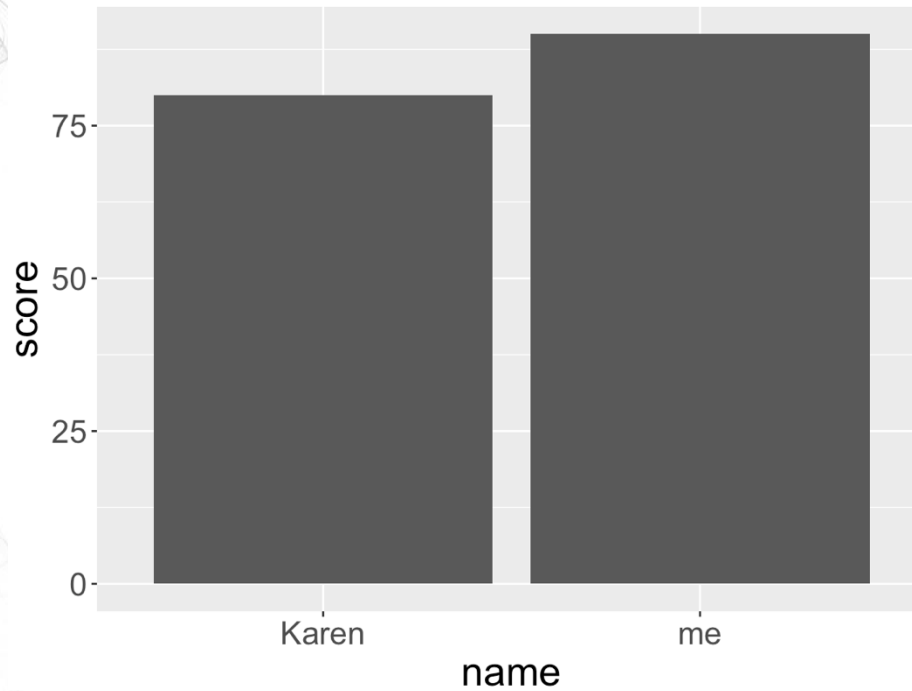
Provide labels and numbers

From: Luo et al., 2020

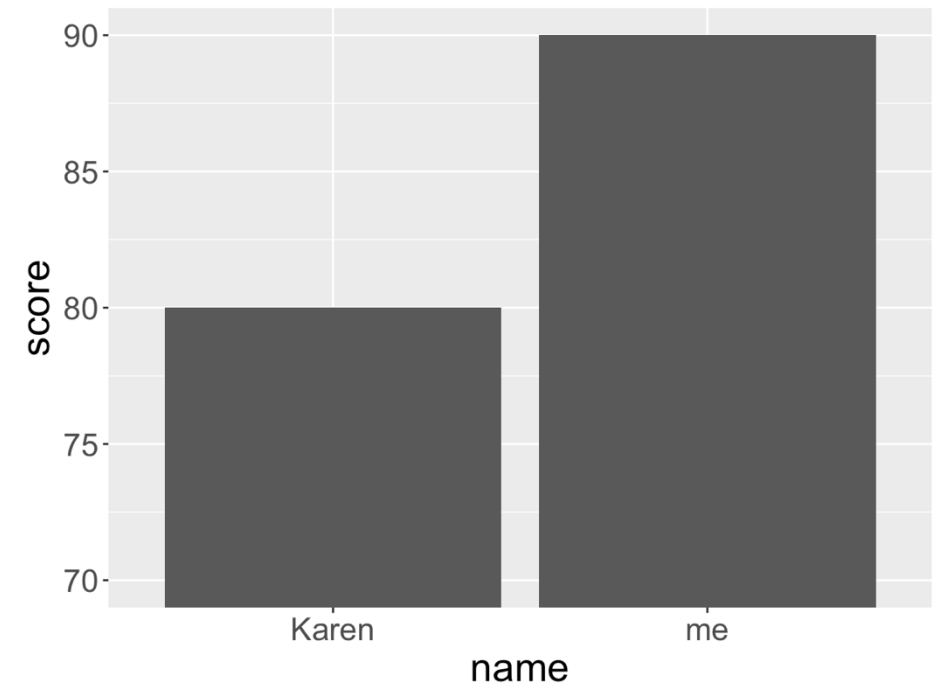
<https://www.nature.com/articles/s42256-020-0167-4/figures/3>

Know where to put 0

```
> df <- data.frame(name=c("me", "Karen"),  
+                  score=c(90,80))  
> ggplot(data=df, aes(x=name, y=score)) +  
+ geom_col() + theme(text = element_text(size=20))
```

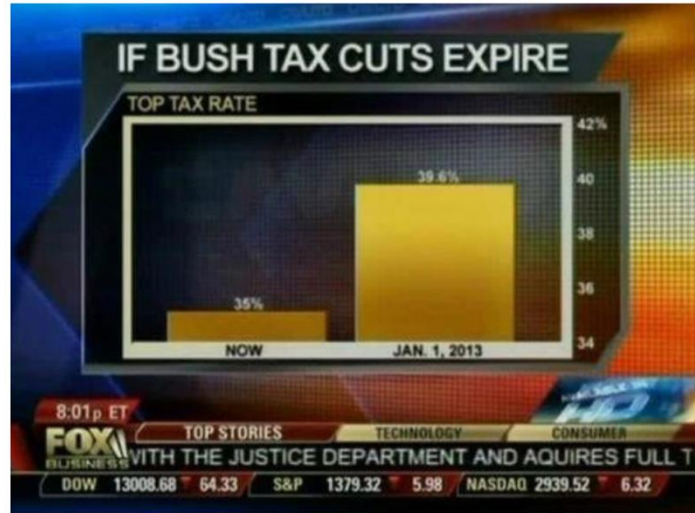


```
> ggplot(data=df, aes(x=name, y=score)) +  
+ geom_col() + theme(text = element_text(size=20)) +  
+ coord_cartesian(ylim=c(70, 90))
```

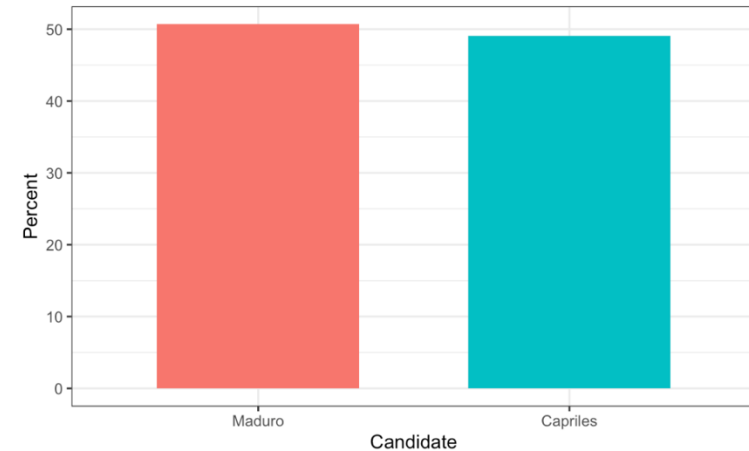
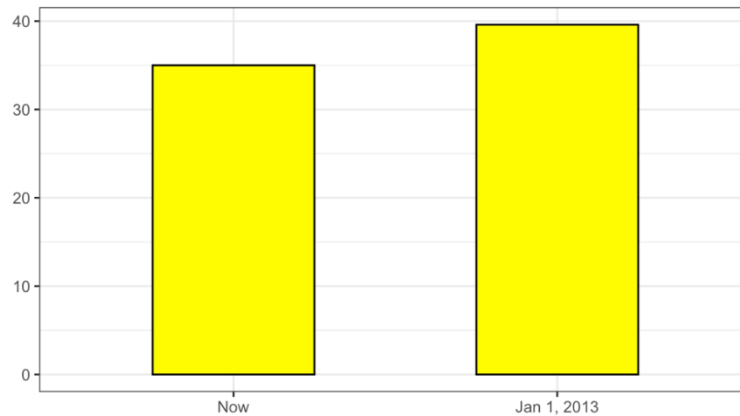


See also: <https://medium.com/mind-talk/what-happens-when-bar-charts-dont-start-with-zero-7db04221417e>

Know where to put 0



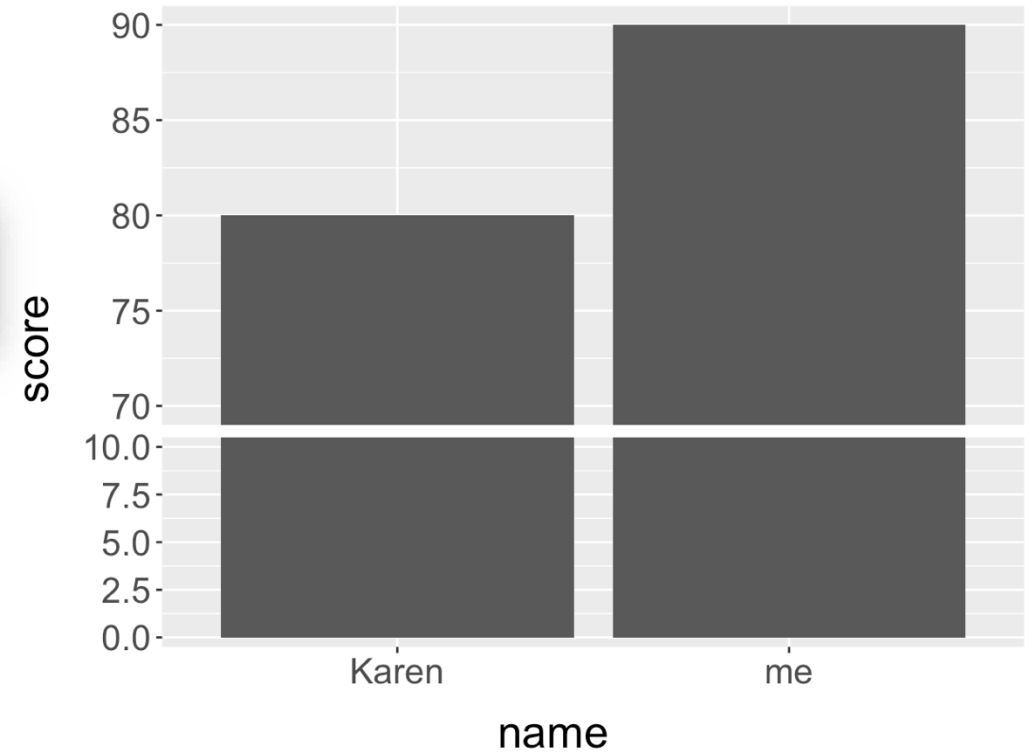
Top Tax Rate If Bush Tax Cut Expires



Know where to put 0

Use the `ggbreak` package

```
> library(ggbreak)
> ggplot(data=df, aes(x=name, y=score)) +
+ geom_col() + theme(text = element_text(size=20)) +
+ coord_cartesian(ylim=c(0, 90)) + scale_y_break(c(10,70))
```

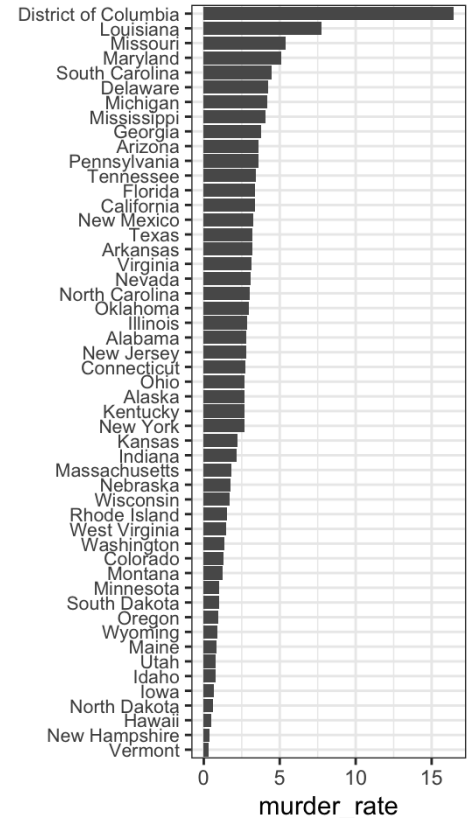
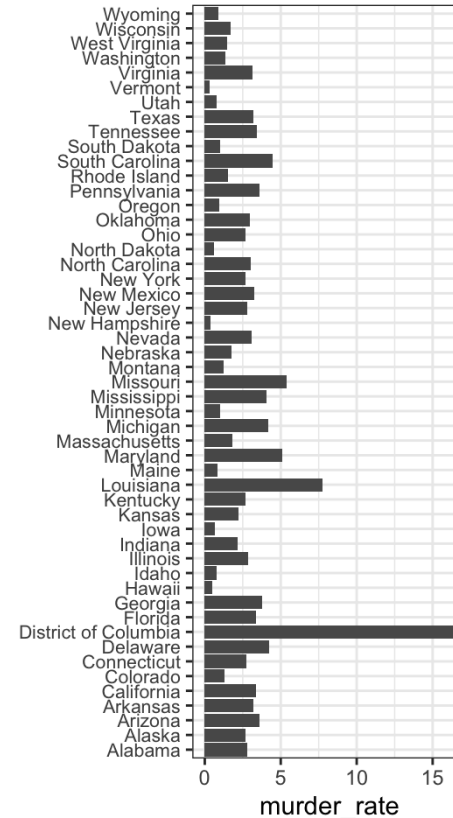


Order your categories meaningfully

```
> data(murders)
> murders |> mutate(murder_rate = total / population * 100000) |>
+ mutate(state = reorder(state, murder_rate)) |>
+ ggplot(aes(state, murder_rate)) +
+ geom_bar(stat="identity") +
+ coord_flip() +
+ theme(axis.text.y = element_text(size = 6)) +
+ xlab("")
```

Use the `reorder` function to arrange your categorical variables.

In the two plots on the right, which one makes it easier to spot the most dangerous and safest states?

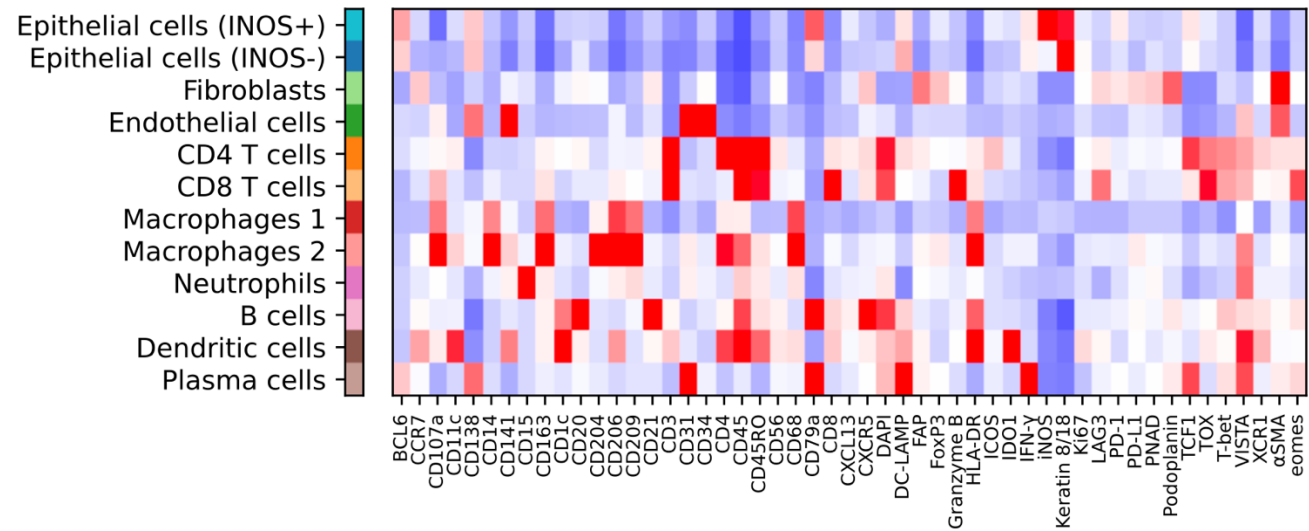
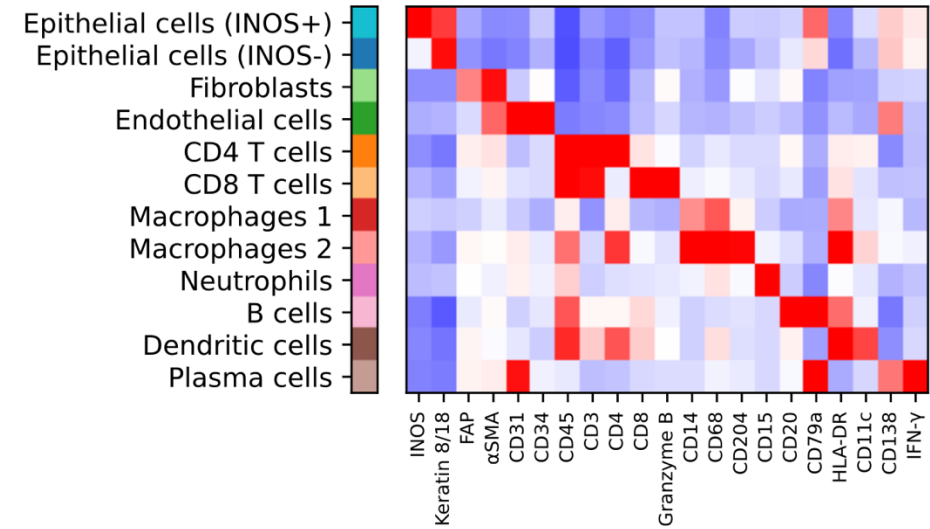


Order your categories meaningfully

The two plots here showed protein expression in different types of cells:

- **Red: enriched**
- **Blue: depleted**

Which one makes it easier to spot the signature proteins for each cell type?

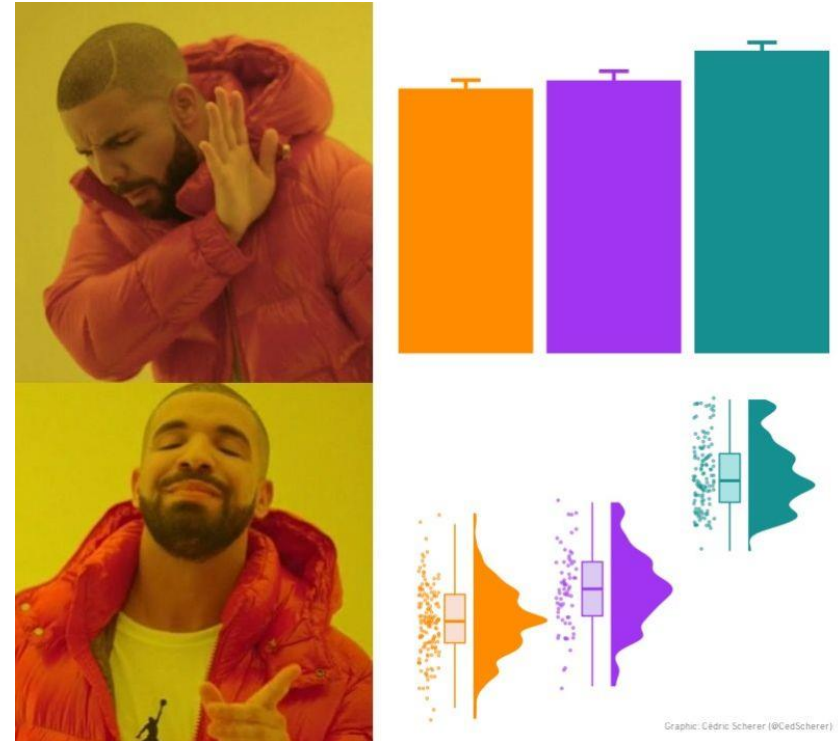


Show your data

Dynamite plot: bar plot with error bar, suggesting that it summarizes multiple numbers.

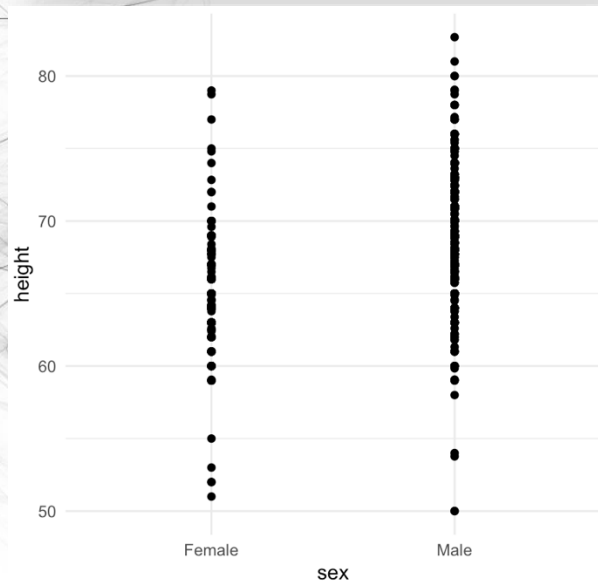
To visualize multiple numbers:

- **Dynamite plot:** mean + SD
- **Box plot:** quartiles, minimum/maximum, outliers
- **Violin/density plot:** full distribution
- **Scatter/swarm plot:** all data points

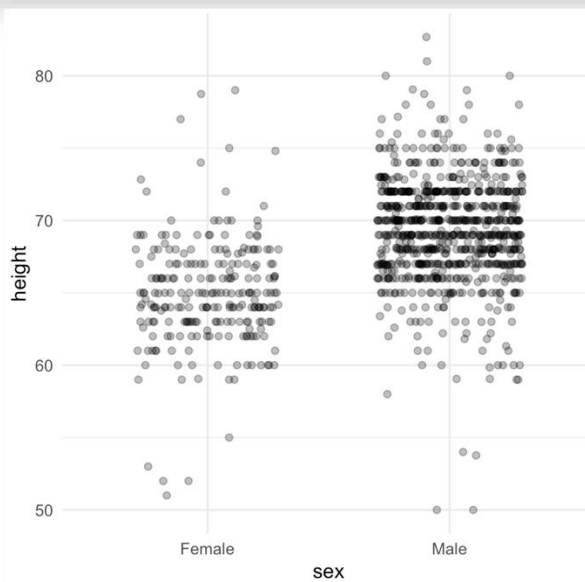


Show your data

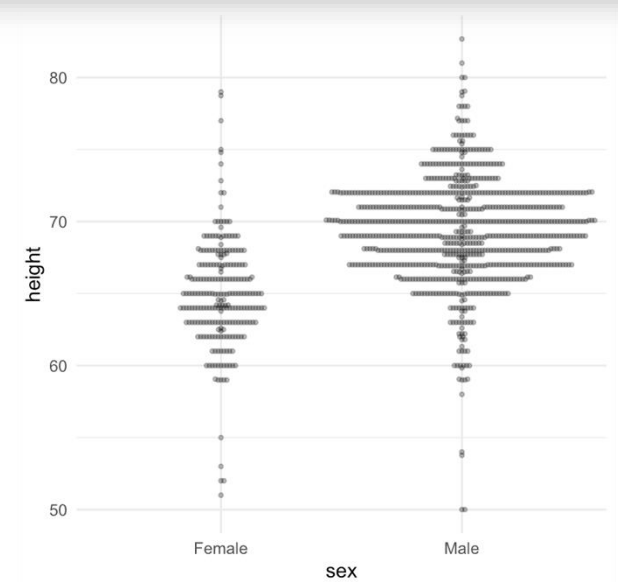
```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_point() +  
+   theme_minimal()
```



```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_jitter(width=0.3, alpha=0.3) +  
+   theme_minimal()
```

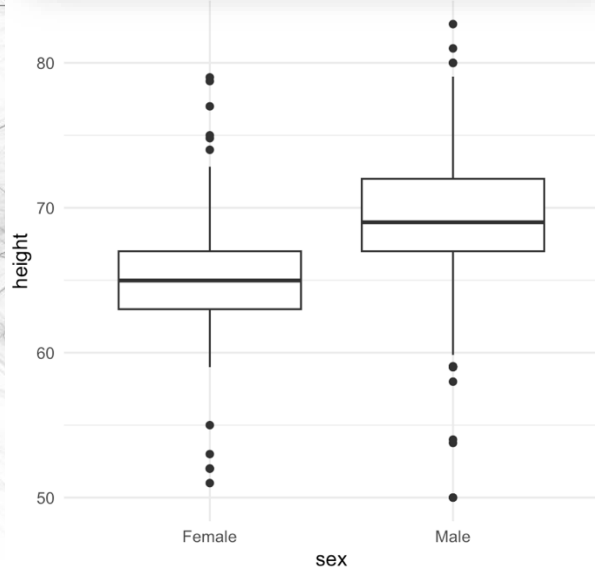


```
> library(ggbeeswarm)  
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_beeswarm(alpha=0.3, cex=0.6, size=0.8) +  
+   theme_minimal()
```

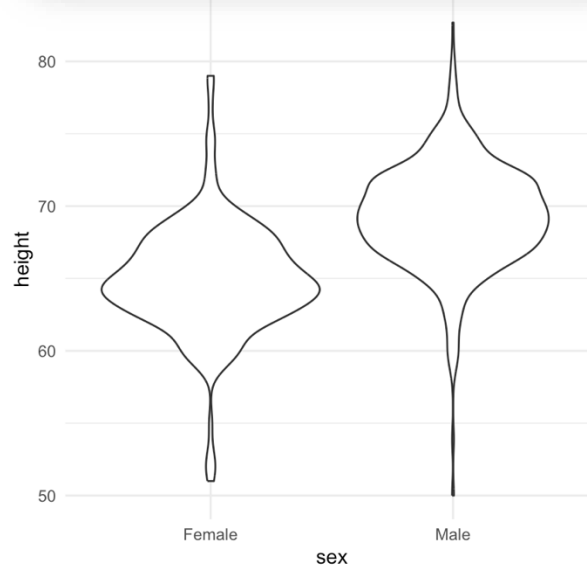


Show your data

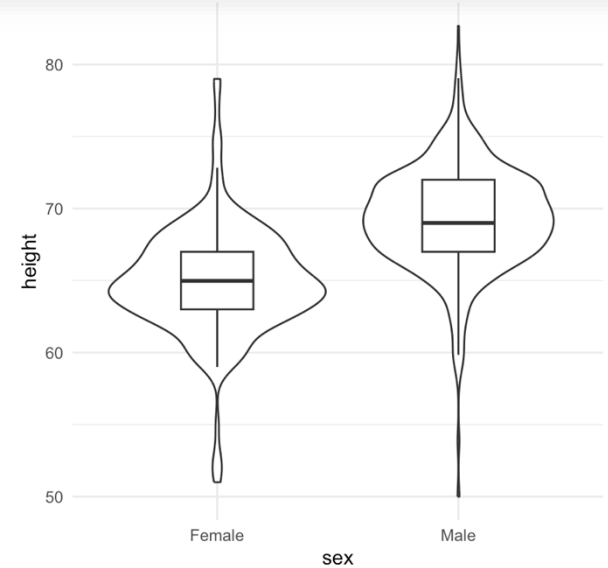
```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_boxplot() +  
+   theme_minimal()
```



```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_violin() +  
+   theme_minimal()
```

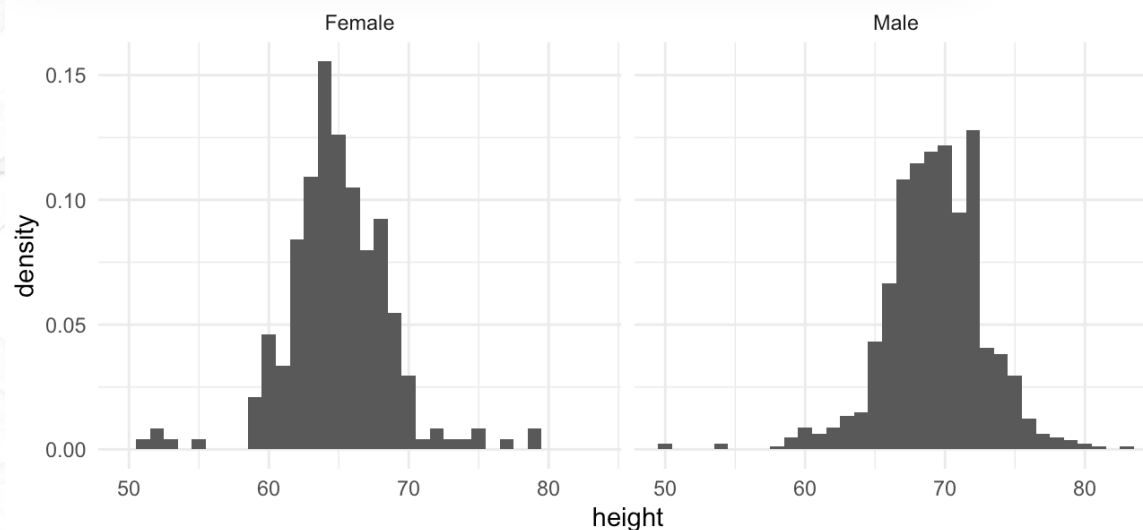


```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_violin() +  
+   geom_boxplot(width=0.3, outliers = FALSE) +  
+   theme_minimal()
```



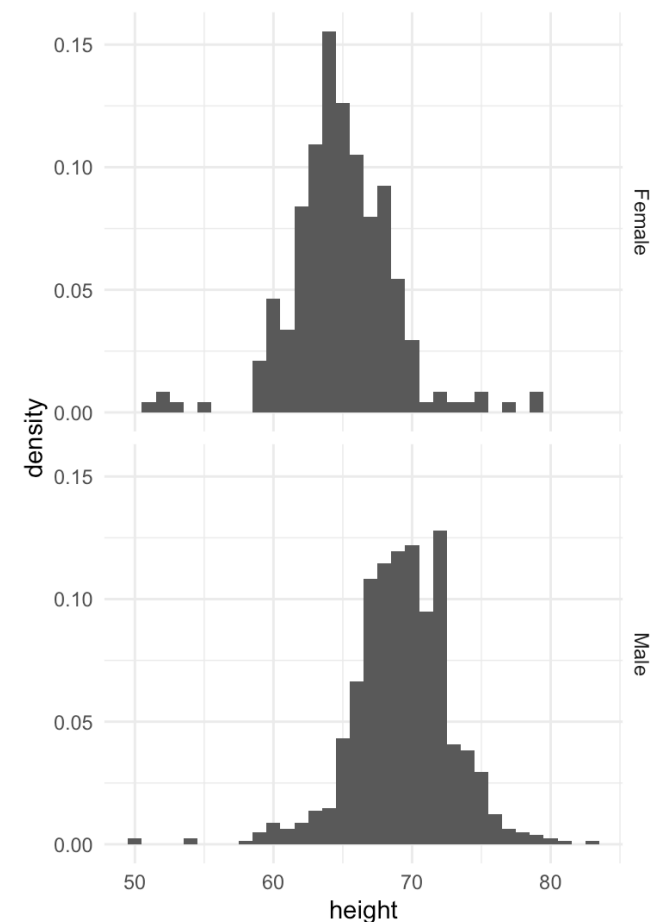
Ease comparison

```
heights |>
  ggplot(aes(height, label=sex)) +
  geom_histogram(aes(y=after_stat(density)), binwidth=1) +
  facet_grid(. ~ sex) +
  theme_minimal()
```



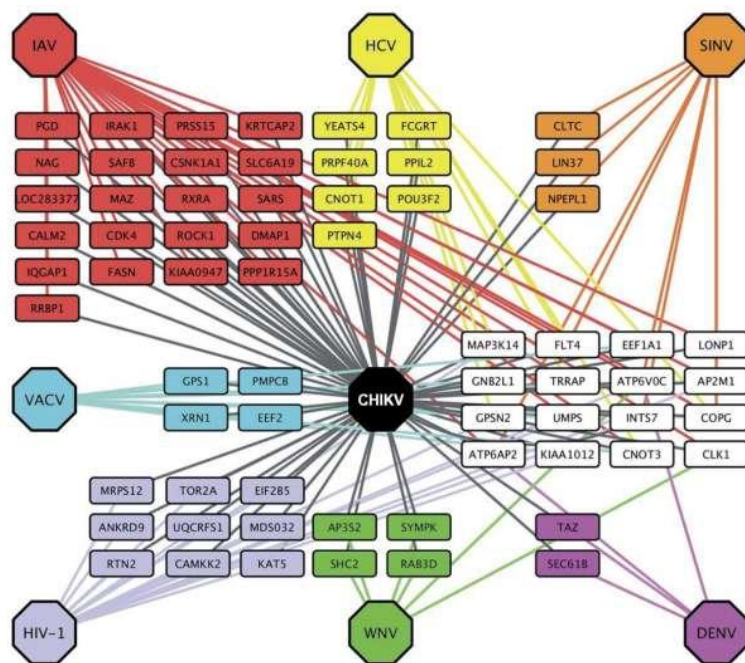
What are we comparing?
Height, so use that as common axes

```
heights |>
  ggplot(aes(height, label=sex)) +
  geom_histogram(aes(y=after_stat(density)), binwidth=1) +
  facet_grid(sex ~ .) +
  theme_minimal()
```



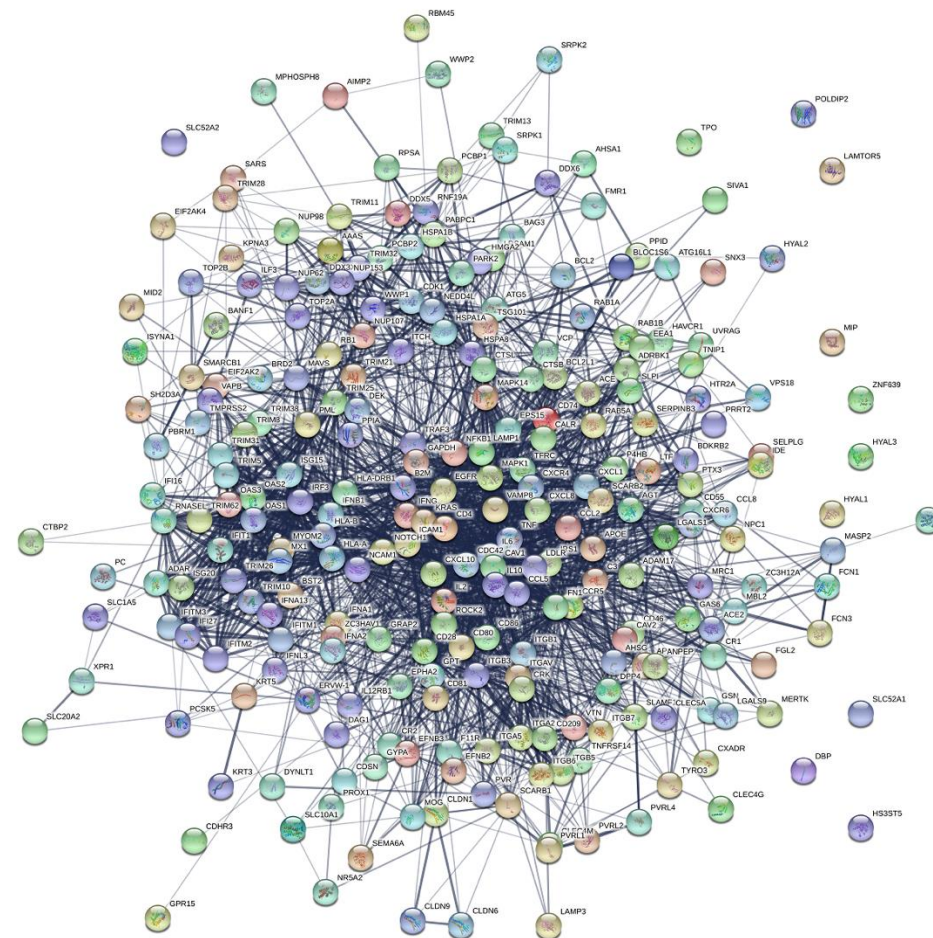
The more, the better?

With a correct amount of information,
you are willing to look into the details



[Treatments for coronavirus: Repurposing existing drugs \(medicalxpress.com\)](https://www.medicalxpress.com)

Look! We have a lot of information here.



[COVID-19: A Drug Repurposing and Biomarker Identification by Using Comprehensive Gene-Disease Associations through Protein-Protein Interaction Network Analysis\[v1\] | Preprints](#)

Significant digits

- Show more significant digits only when necessary: e.g., highlighting minor differences

state	year	Measles	Pertussis	Polio
California	1940	37.8826320	18.3397861	0.8266512
California	1950	13.9124205	4.7467350	1.9742639
California	1960	14.1386471	NA	0.2640419
California	1970	0.9767889	NA	NA
California	1980	0.3743467	0.0515466	NA

state	year	Measles	Pertussis	Polio
California	1940	37.9	18.3	0.8
California	1950	13.9	4.7	2.0
California	1960	14.1	NA	0.3
California	1970	1.0	NA	NA
California	1980	0.4	0.1	NA

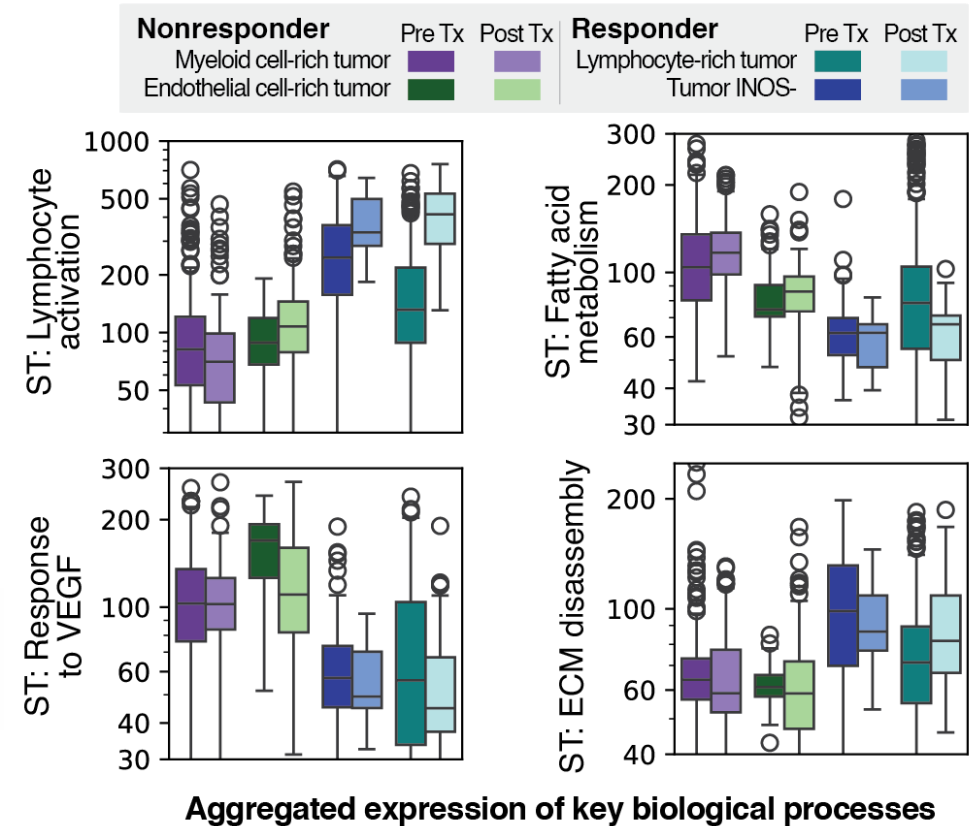
Use round

```
> array <- c(1, 1.143523453451432324, 2.3, 3.56)
> round(array, 2)
[1] 1.00 1.14 2.30 3.56
```

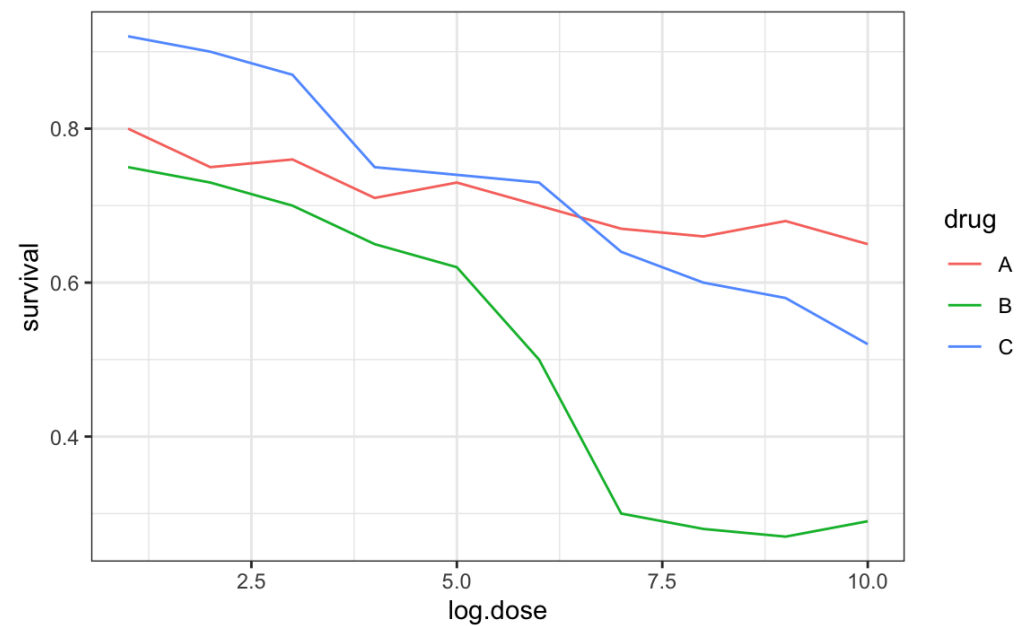
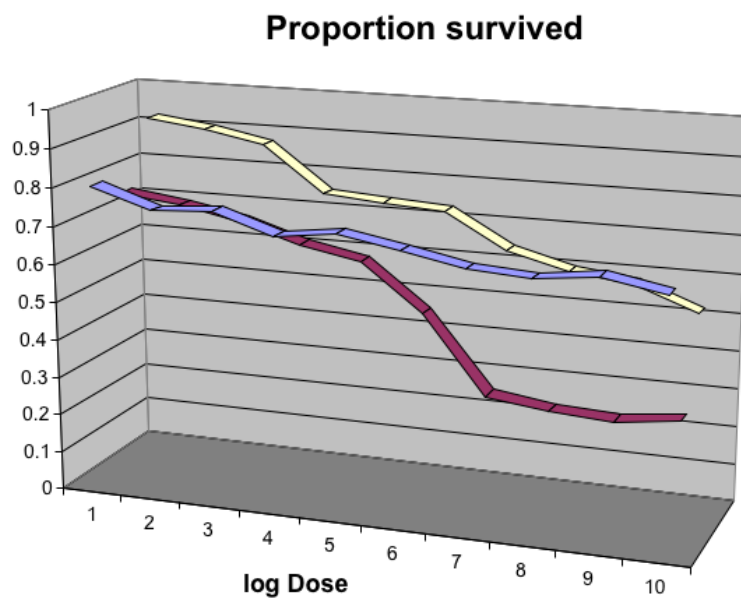

Make good use of colors

- Use no more than six colors in a plot
- Hue versus shade
- Be consistent between plots (use the same palette)
- Use color-blind friendly colors

```
> color_blind_friendly_cols <-  
  c("#999999", "#E69F00", "#56B4E9", "#009E73",  
    "#F0E442", "#0072B2", "#D55E00", "#CC79A7")
```

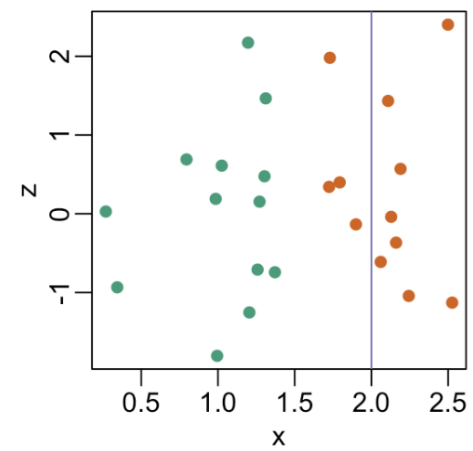
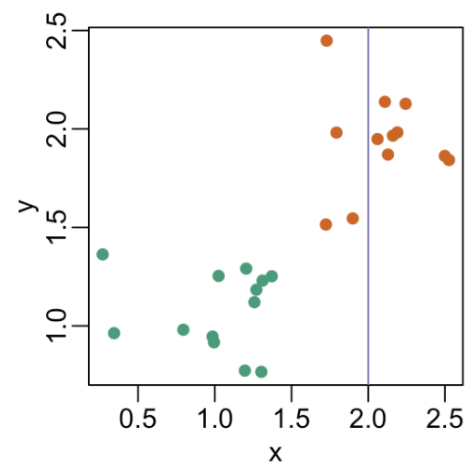
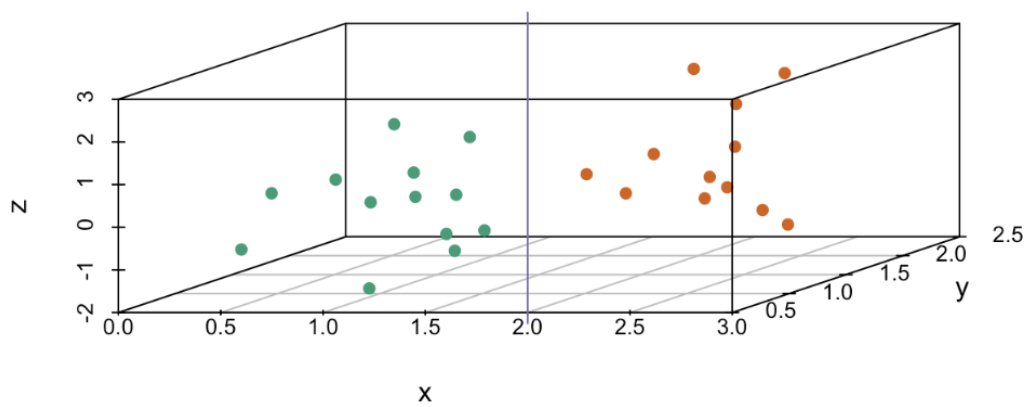


Avoid (pseudo) 3D plots



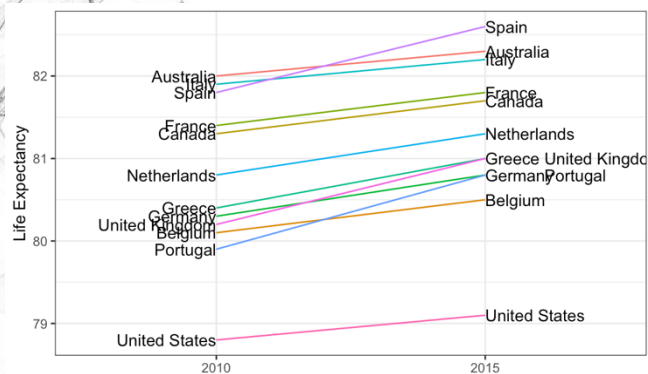
Just don't use pseudo 3D!

Avoid (pseudo) 3D plots

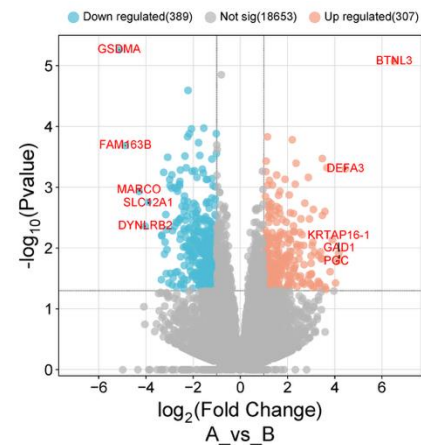


Take note of interesting new plots in your daily life

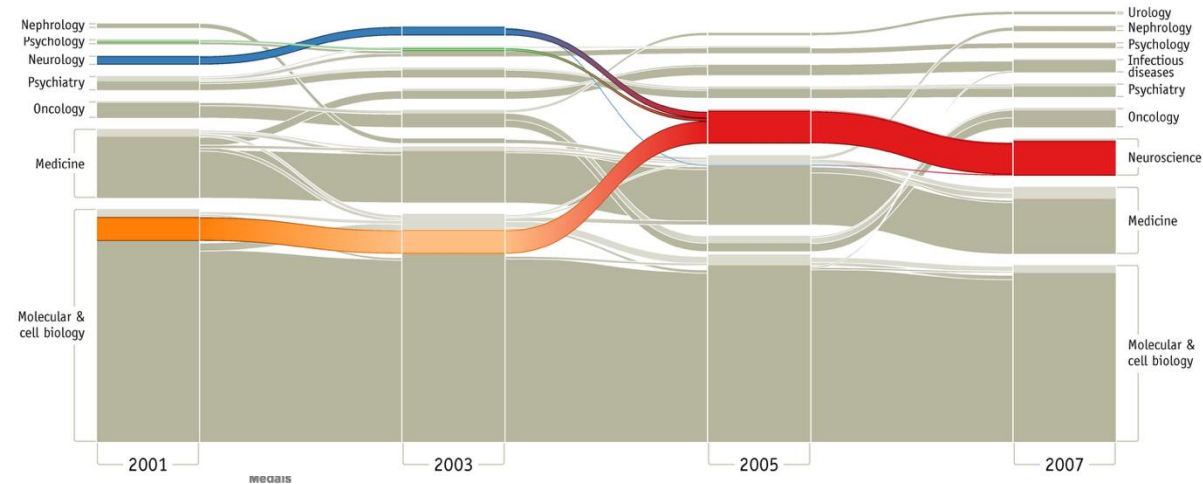
Slope chart



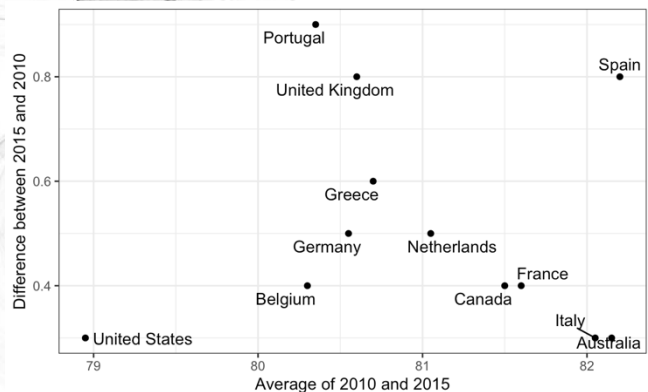
Volcano plot



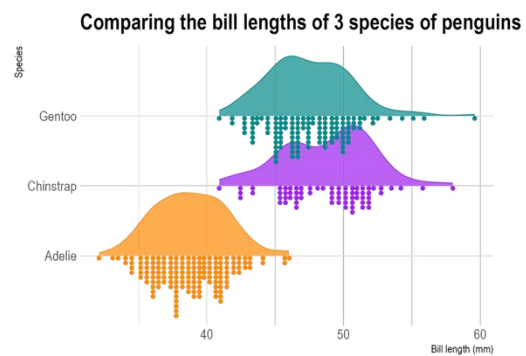
Alluvial diagram



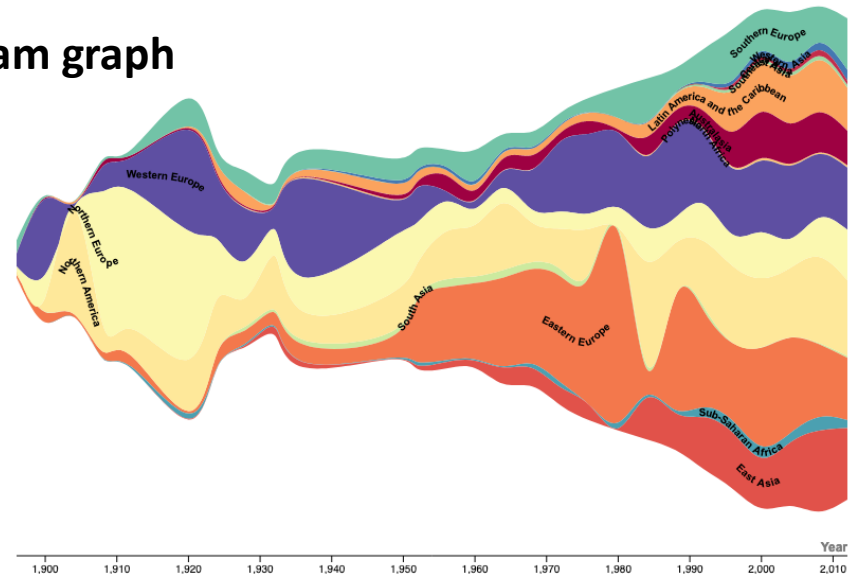
Bland-Altman plot



Raincloud plot



Stream graph





In this lecture

- How to do a good visualization? /
How to avoid bad visualization?
- **Case study: vaccines and measles**

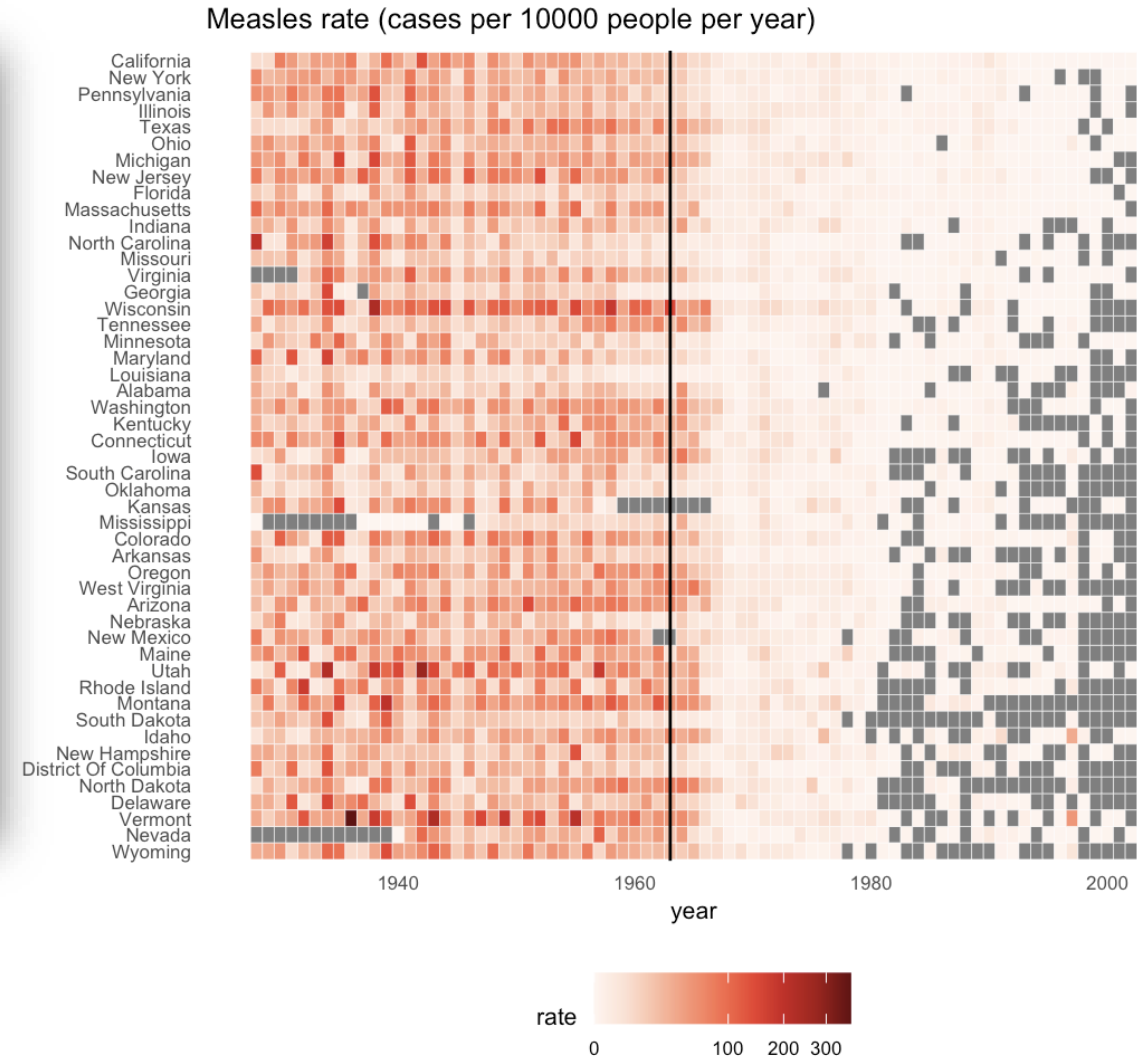
Case study: vaccines and infectious diseases

```
library(tidyverse)
library(dslabs)
library(RColorBrewer)

data(us_contagious_diseases)

dat <- us_contagious_diseases |>
  filter(!(state %in% c("Hawaii", "Alaska")), # Excluding due to insufficient data
         disease == "Measles") |>
  mutate(rate = count / population * 10000 * (52 / weeks_reporting)) |>
  mutate(state = reorder(state, population, FUN = median, na.rm = TRUE))

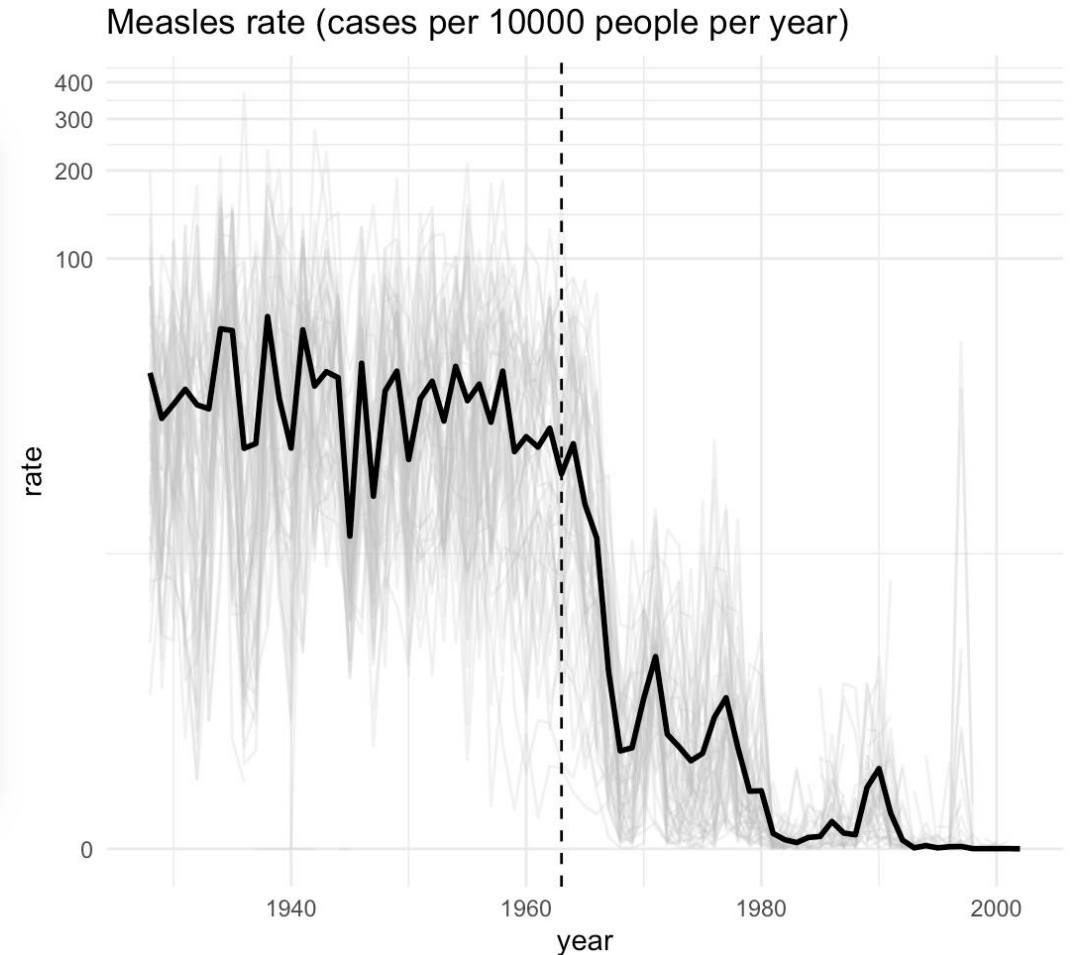
dat |>
  ggplot(aes(year, state, fill=rate)) +
  geom_tile(color = "white") +
  scale_x_continuous() +
  scale_fill_gradientn(colors = brewer.pal(9, "Reds"), trans = "sqrt") +
  geom_vline(xintercept = 1963, col = "black") +
  theme_minimal() +
  theme(panel.grid = element_blank(),
        legend.position = "bottom",
        text = element_text(size = 8)) +
  labs(title = "Measles rate (cases per 10000 people per year)", y = "")
```



Case study: vaccines and infectious diseases

```
avg <- us_contagious_diseases |>
  filter(disease == "Measles") |>
  group_by(year) |>
  summarize(rate = sum(count, na.rm = TRUE) /
             sum(population, na.rm = TRUE) * 10000)

dat |>
  filter(!is.na(rate)) |>
  ggplot() +
  geom_line(aes(year, rate, group=state),
            data = dat, color = "grey",
            alpha = 0.2, na.rm = TRUE) +
  geom_line(aes(year, rate),
            data = avg, color = "black",
            size = 1) +
  geom_vline(xintercept = 1963, col = "black", lty = 2) +
  scale_y_continuous(transform = "log1p") +
  theme_minimal() +
  labs(title = "Measles rate (cases per 10000 people per year)")
```





In this lecture

- How to do a good visualization? / How to avoid bad visualization?
 - Effectiveness of different visual cues
 - Place your 0s
 - Order your categories
 - Show your data
 - Remove unnecessary stuff
 - No 3D, please
- Case study: vaccines and measles



Next class is a lab

- Bring your own laptop, please make sure RStudio is installed and functional on your laptop.
- You will be given a lab sheet, you are expected to answer the questions in the sheet in class.
 - You are encouraged to try different code or arguments.
 - You can discuss with other students for ideas (DO NOT copy)
 - For questions, ask Google, GPT, the two TAs, or me
- Sample answers will be given after the class